

Introduction to Artificial Intelligence

Lecture 0: Artificial Intelligence

Prof. Gilles Louppe

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ChatGPT PLUS

Help me pick

a gift for my dad who loves fishing

Brainstorm edge cases

for a function with birthdate as input, horoscope as ou...

Make up a story

about Sharky, a tooth-brushing shark superhero

Create a personal webpage for me

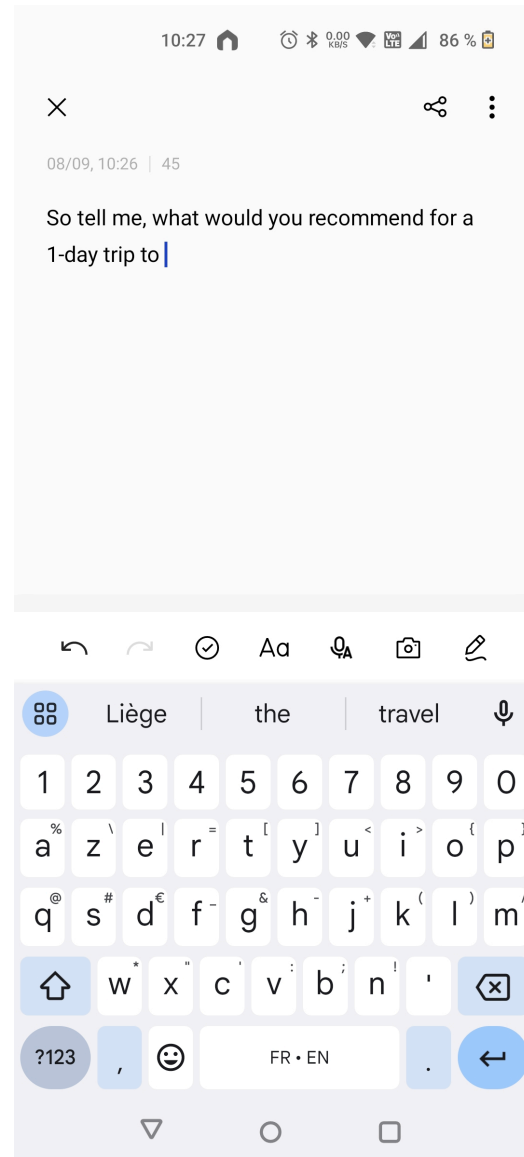
after asking me three questions

 Send a message



One simple idea:

Guess the next word



In the 1960s, Armstrong _____

In the 1960s, Armstrong performed ____

In the 1960s, Armstrong performed a moonwalk ____

In the 1960s, Armstrong performed a moonwalk on the ____

In the 1960s, Armstrong performed a moonwalk on the lunar ____

In the 1960s, Armstrong performed a moonwalk on the lunar surface and said ____



What does it mean to predict the next token well enough? It's actually [...] a deeper question than it seems. Predicting the next token well means that you understand the underlying reality that led to the creation of that token.

— Ilya Sutskever, 2023

This explains why a model that only guesses the next word ...

- gives different answers to the same question;
- cannot count, compute, or reason;
- invents things and cannot cite sources;
- can hardly correct its own mistakes once they have been made.

How do we go from here to agents that use computers and prove theorems?

Artificial Intelligence

A definition of AI?

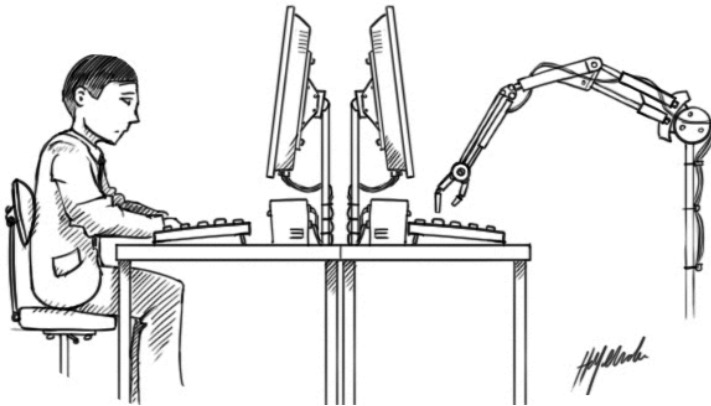


Artificial intelligence is the science of making machines do things that would require intelligence if done by men.

— Marvin Minsky, 1968

The Turing test

A computer passes the **Turing test** (aka the Imitation Game) if a human operator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.



*Can machines think?
(Alan Turing, 1950)*

In 2025, a language model was judged to be the human **73%** of the time, more often than the humans themselves.



Aeronautical engineering texts do not define the goal of their field as making "machines that fly so exactly like pigeons that they can fool even other pigeons."

— Russell and Norvig

A modern definition of AI

'AI system' means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.

— EU AI Act, Article 3, 2024

A short history of AI

1940-1950: Early days

- 1943: McCulloch and Pitts: Boolean circuit model of the brain.
- 1950: Turing's "Computing machinery and intelligence".

1950-1970: Excitement and expectations

- 1950s: Early AI programs, including Samuel's checkers program, Newell and Simon's Logic Theorist and Gelernter's Geometry Engine.
- 1956: Dartmouth meeting: "Artificial Intelligence" adopted.
- 1958: Rosenblatt invents the perceptron.
- 1965: Robinson's complete algorithm for logical reasoning.
- 1966-1974: AI discovers computational complexity.



The Dartmouth workshop (1956)

The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.



1970-1990: Knowledge-based approaches

- 1969: Neural network research almost disappears after Minsky and Papert's book (1st AI winter).
- 1969-1979: Early development of knowledge-based systems.
- 1980-1988: Expert systems industrial boom.
- 1988-1993: Expert systems industry busts (2nd AI winter).

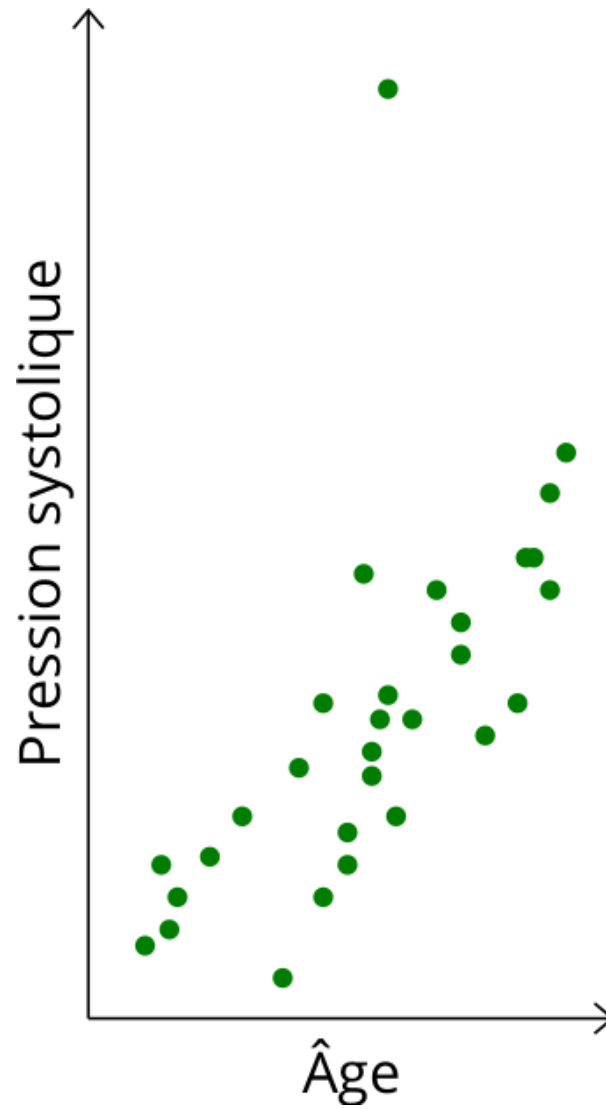
1990-2022: Statistical approaches

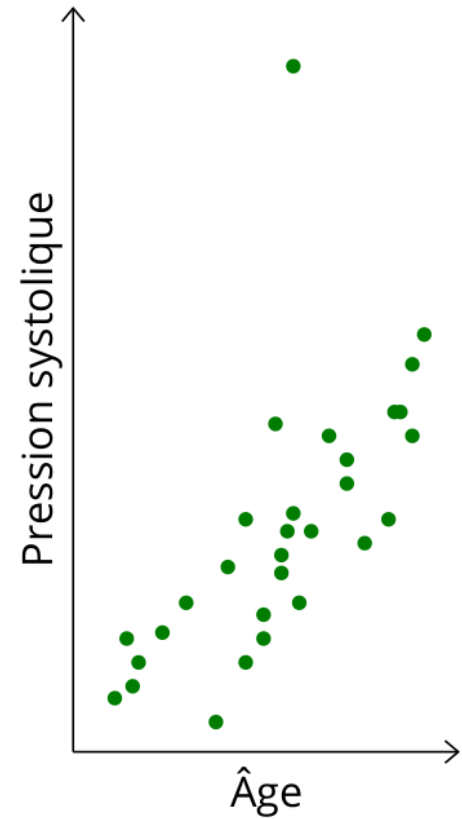
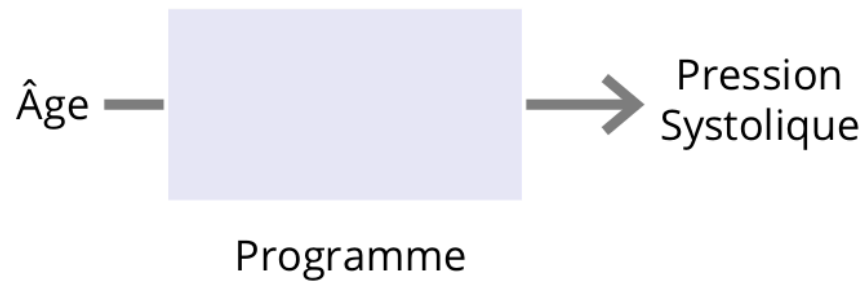
- 1985-1995: The return of neural networks.
- 1988-: Resurgence of probability, focus on uncertainty, general increase in technical depth.
- 1995-2010: New fade of neural networks.
- 2000-: Availability of very large datasets.
- 2010-: Availability of fast commodity hardware (GPUs).
- 2012-: Resurgence of neural networks with deep learning approaches.
- 2017: Attention is all you need (transformers).

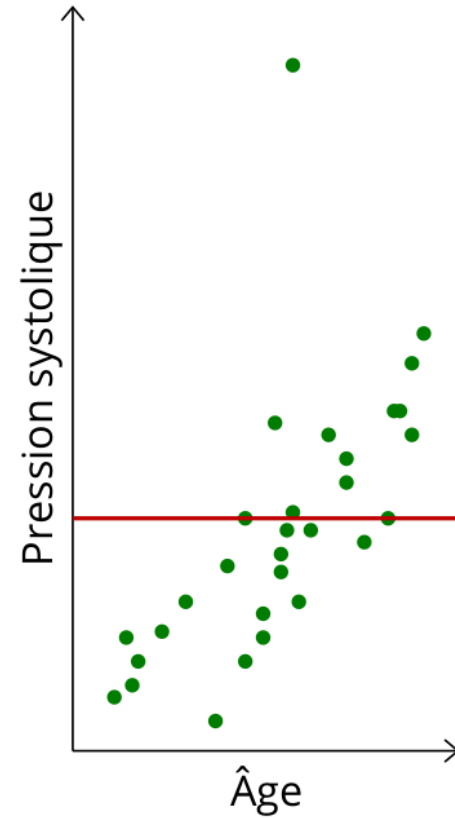
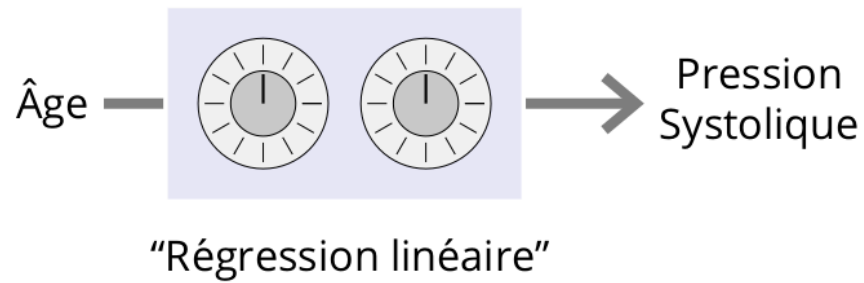
2022-Present: From assistants to agents

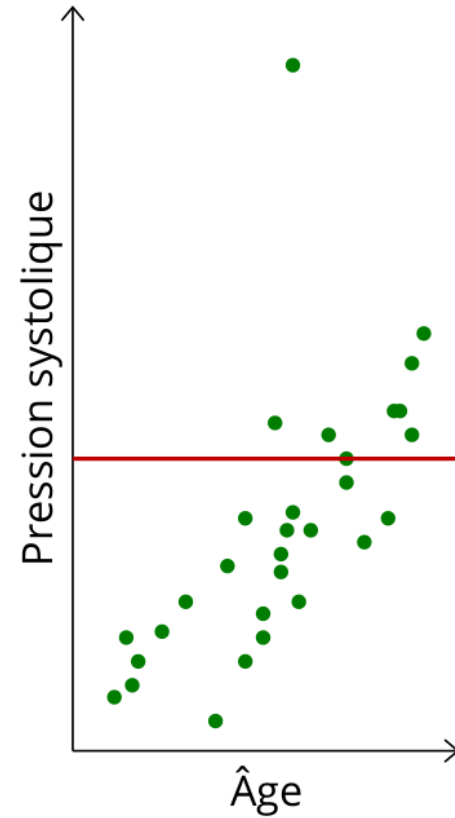
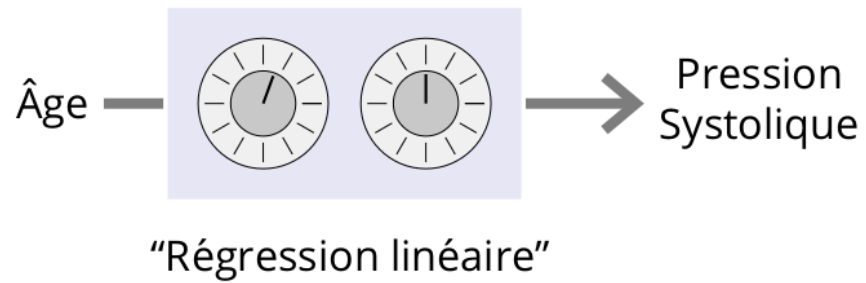
- 2022: ChatGPT released to the public.
- 2024: Reasoning models write before answering.
- 2024: Nobel Prizes in Physics (Hopfield, Hinton) and Chemistry (Baker, Hassabis, Jumper).
- 2025: Barto and Sutton receive the Turing Award for reinforcement learning.
- 2025: Gold-medal score at the International Mathematical Olympiad.
- 2025-: Agents that write code and use computers.
- 2026: Agents claim a proof for the Navier–Stokes Millennium Problem.

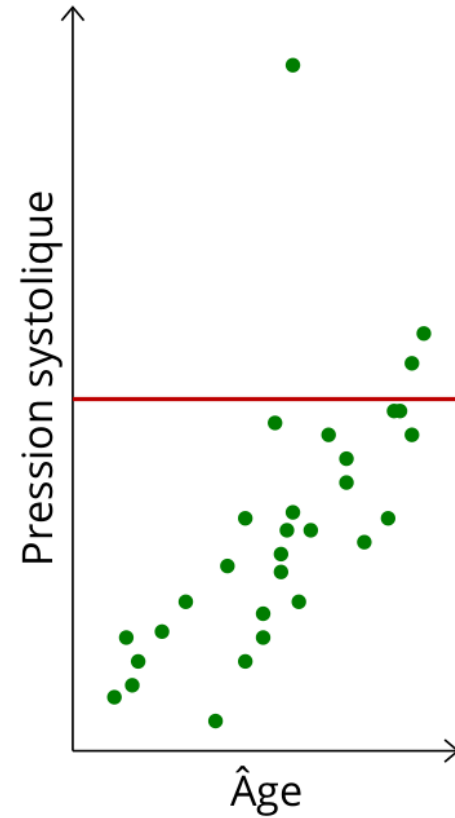
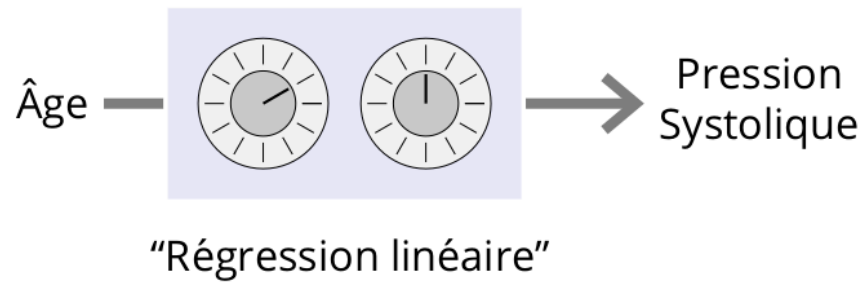
The deep learning revolution

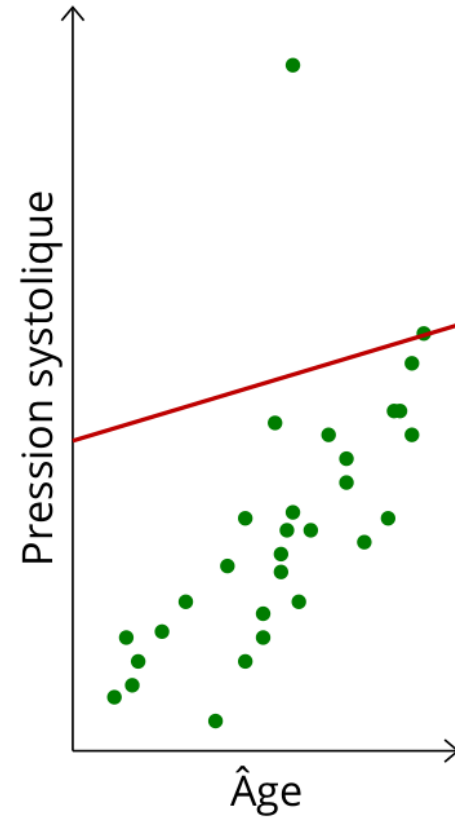
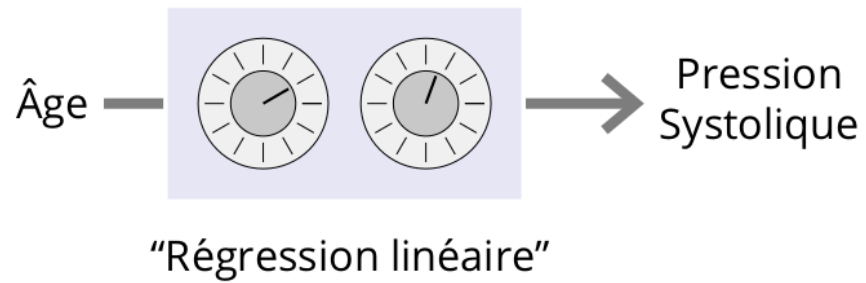


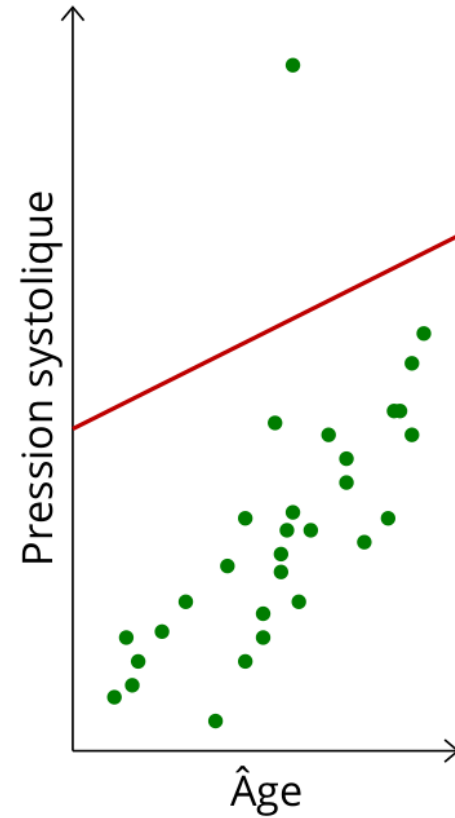
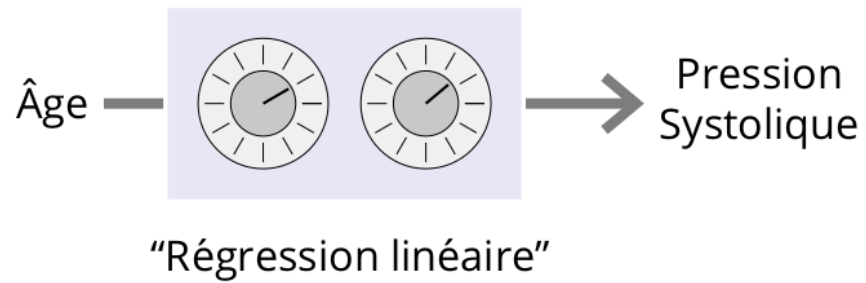


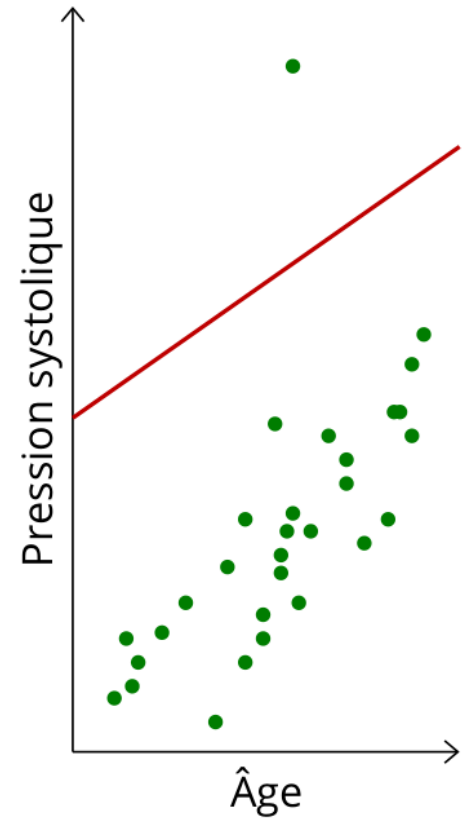
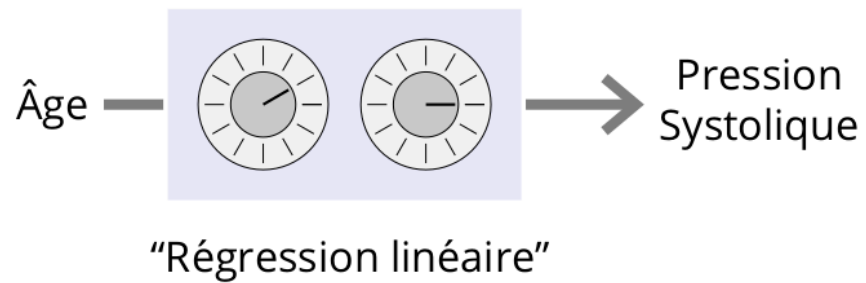


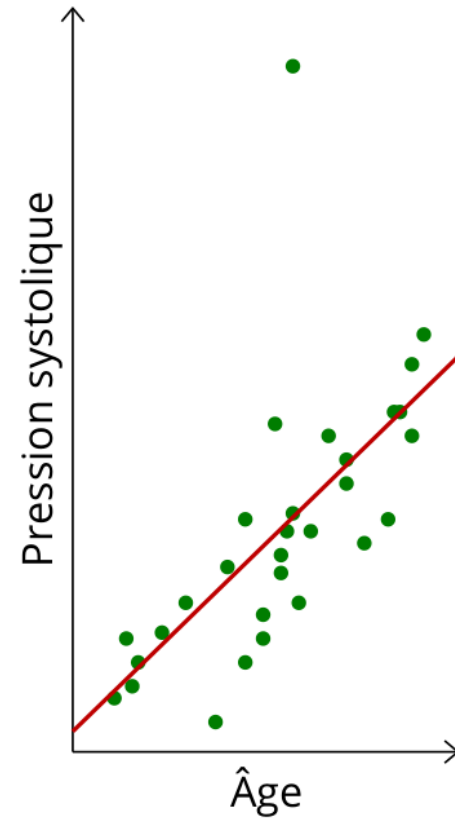
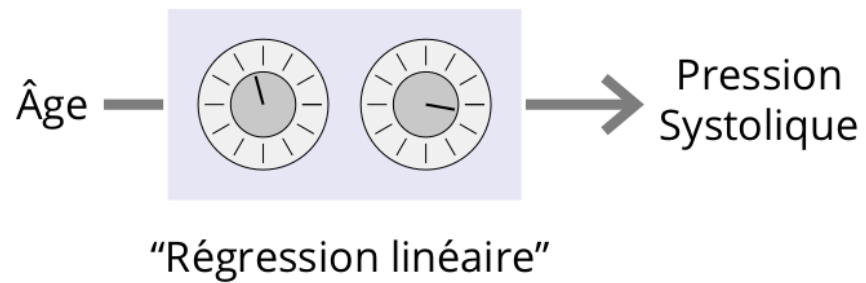






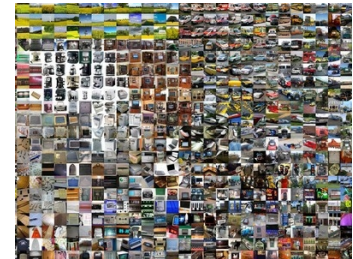
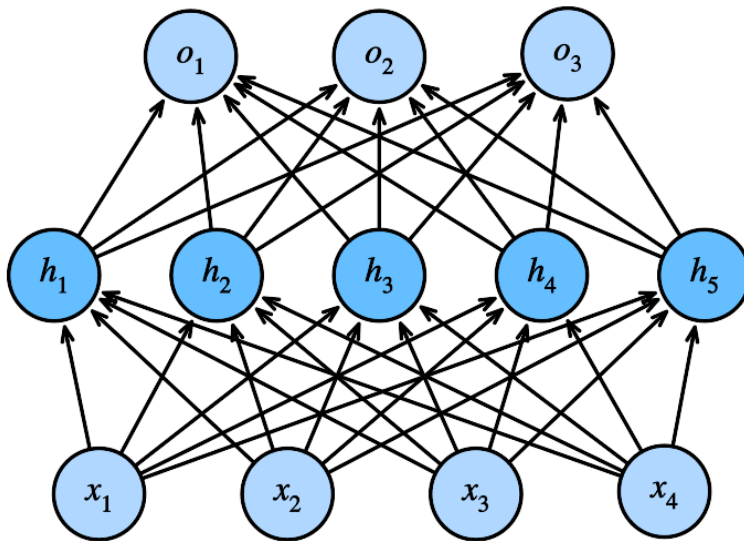




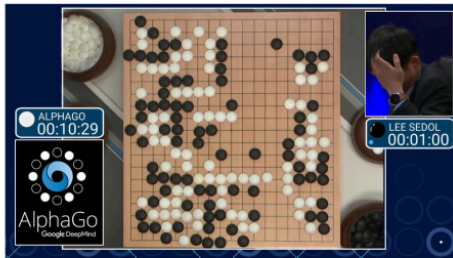
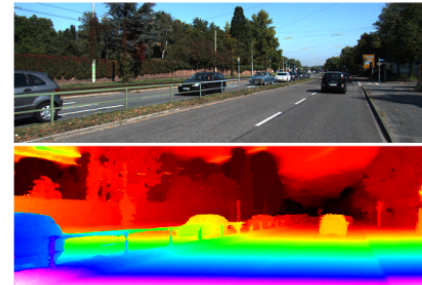
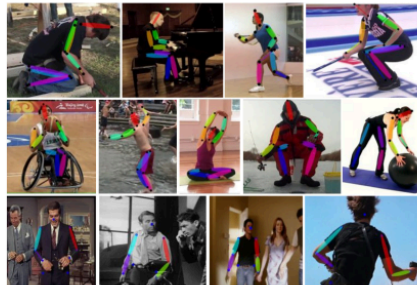
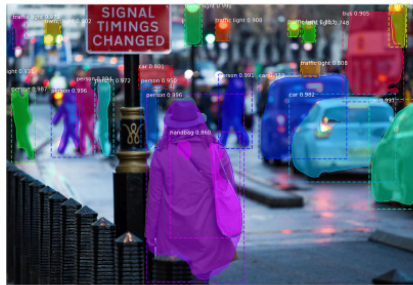


Deep learning **scales up** the statistical and machine learning approaches by

- using larger models known as neural networks,
- training on larger datasets,
- using more compute resources.



Specialized neural networks can be trained to achieve super-human performance on many complex tasks that were previously thought to be out of reach for machines.



A person riding a motorcycle on a dirt road.



Two dogs play in the grass.



A group of young people playing a game of frisbee.



Two hockey players are fighting over the puck.

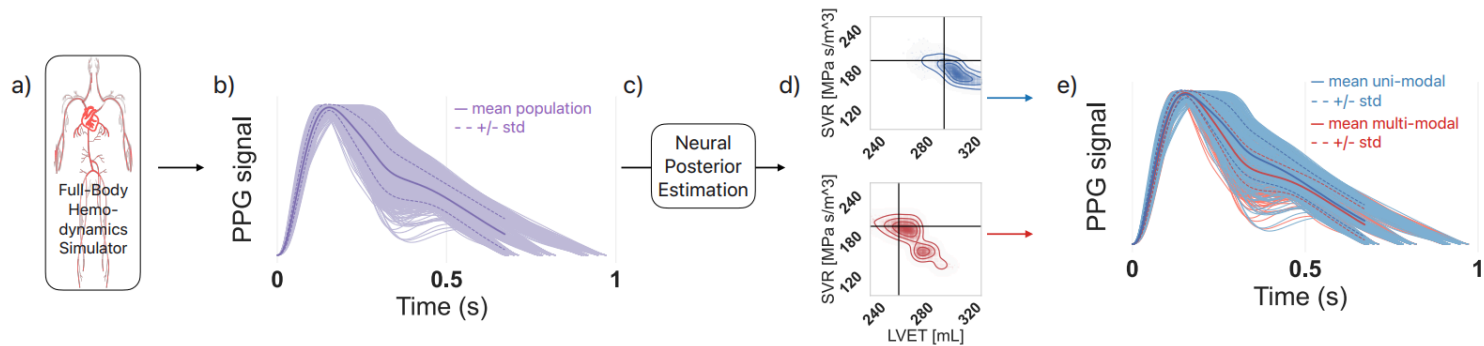
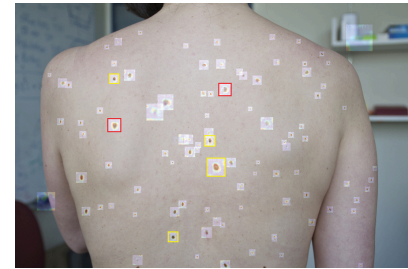
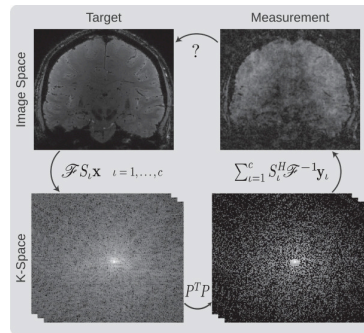
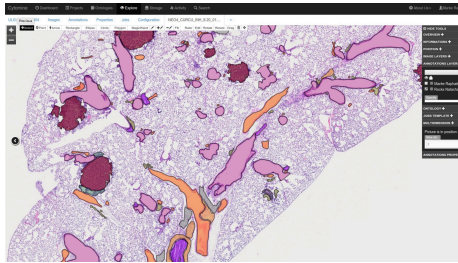


I: Jane went to the hallway.
I: Mary walked to the bathroom.
I: Sandra went to the garden.
I: Daniel went back to the garden.
I: Sandra took the milk there.
Q: Where is the milk?
A: garden

(Top) Scene understanding, pose estimation, geometric reasoning.

(Bottom) Planning, Image captioning, Question answering.

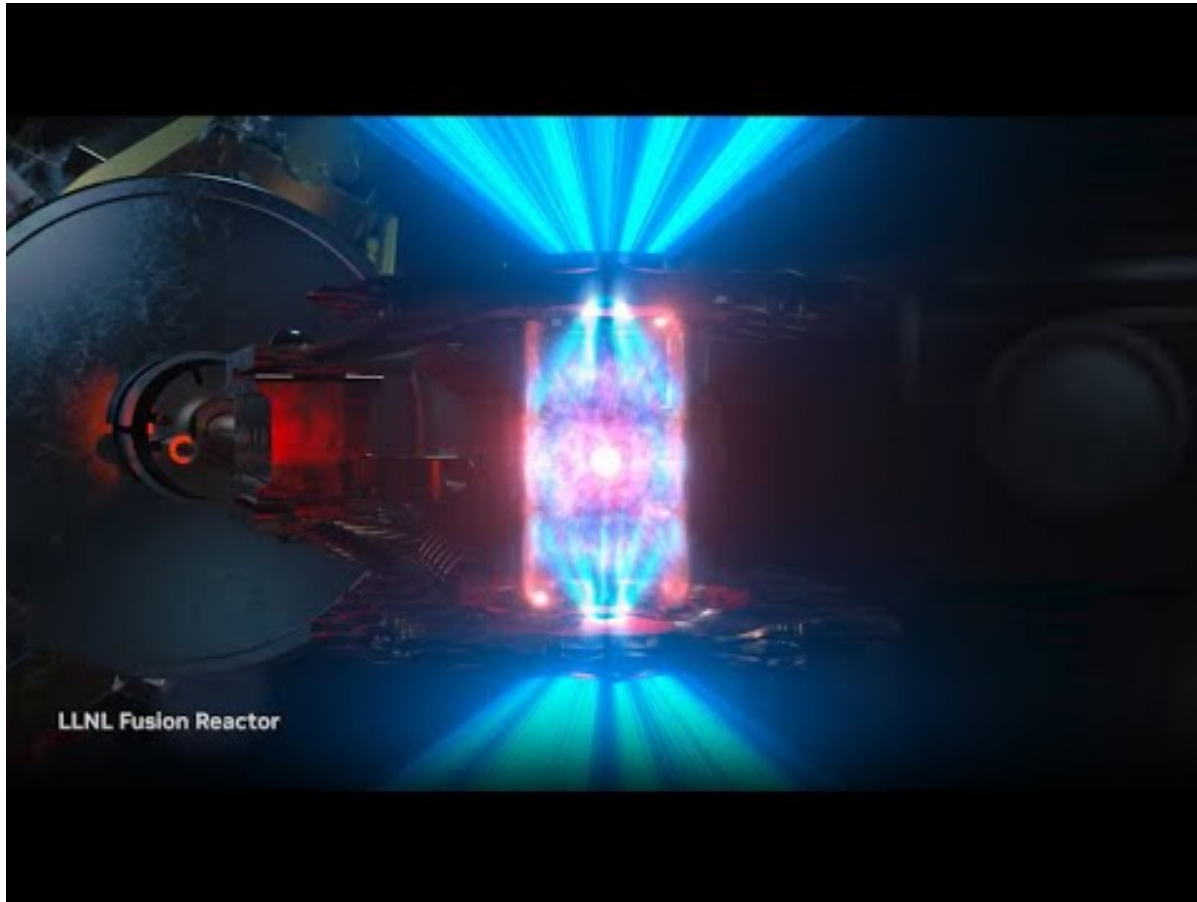
Neural networks form **primitives** that can be transferred to many domains.



(Top) Analysis of histological slides, denoising of MRI images, nevus detection.
 (Bottom) Whole-body hemodynamics reconstruction from PPG signals.



Autonomous cars (Waymo, 2022)



Powering the future of clean energy (NVIDIA, 2023)



How AI is advancing medicine (Google, 2018)

Deep learning can also **solve problems that no one could solve before.**

AlphaFold: From a sequence of amino acids to a 3D structure

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Highly accurate protein structure prediction with AlphaFold

[John Jumper](#) , [Richard Evans](#), [Alexander Pritzel](#), [Tim Green](#), [Michael Figurnov](#), [Olaf Ronneberger](#), [Kathryn Tunyasuvunakool](#), [Russ Bates](#), [Augustin Židek](#), [Anna Potapenko](#), [Alex Bridgland](#), [Clemens Meyer](#), [Simon A. A. Kohl](#), [Andrew J. Ballard](#), [Andrew Cowie](#), [Bernardino Romera-Paredes](#), [Stanislav Nikolov](#), [Rishub Jain](#), [Jonas Adler](#), [Trevor Back](#), [Stig Petersen](#), [David Reiman](#), [Ellen Clancy](#), [Michal Zielinski](#), ... [Demis Hassabis](#) 

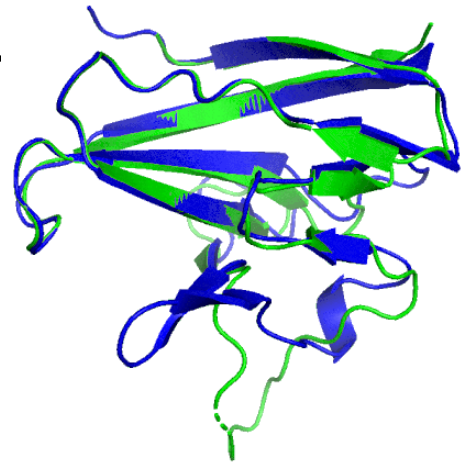
[+ Show authors](#)

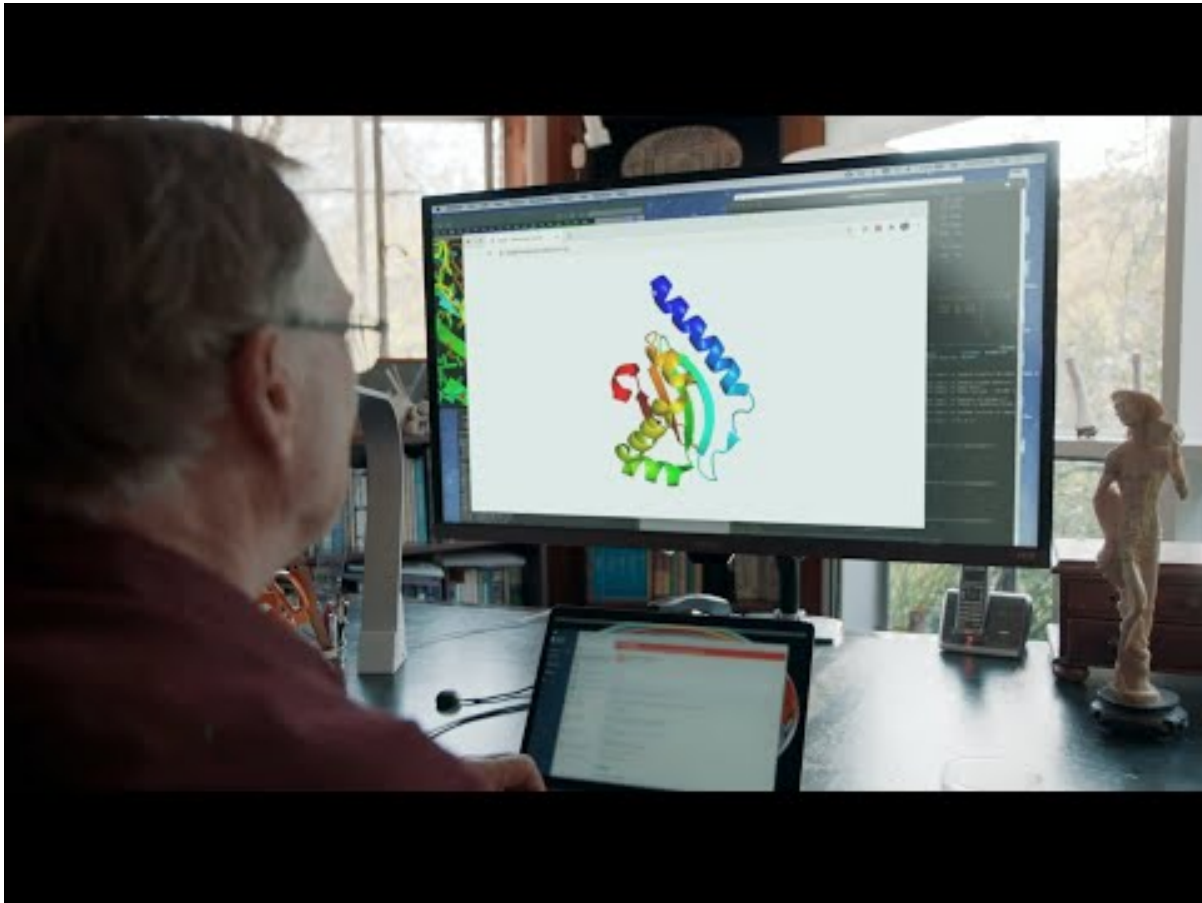
[Nature](#) **596**, 583–589 (2021) | [Cite this article](#)

1.42m Accesses | **12k** Citations | **3493** Altmetric | [Metrics](#)

Abstract

Proteins are essential to life, and understanding their structure can facilitate a mechanistic understanding of their function. Through an enormous experimental effort^{1,2,3,4}, the structures of around 100,000 unique proteins have been determined⁵, but this represents a small fraction of the billions of known protein sequences^{6,7}. Structural coverage is bottlenecked by the months to years of painstaking effort required to determine a single protein structure. Accurate computational approaches are needed to address this gap and to enable large-scale structural bioinformatics. Predicting the three-dimensional structure that a protein will adopt based solely on its amino acid sequence—the structure prediction component of the ‘protein folding problem’⁸—has been an important open research problem for more than 50 years⁹. Despite recent progress^{10,11,12,13,14}, existing methods fall far short of





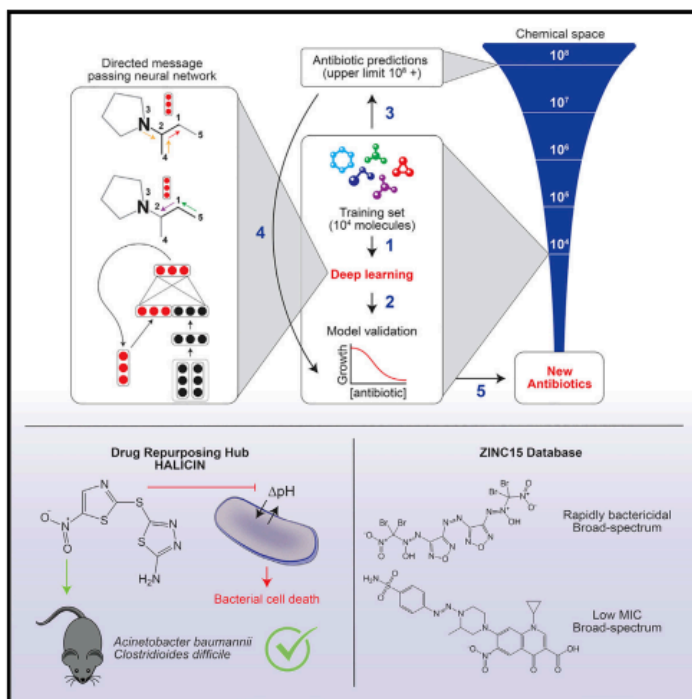
AI for Science (Deepmind, AlphaFold, 2020)

Drug discovery with graph neural networks

Cell

A Deep Learning Approach to Antibiotic Discovery

Graphical Abstract



Authors

Jonathan M. Stokes, Kevin Yang,
Kyle Swanson, ..., Tommi S. Jaakkola,
Regina Barzilay, James J. Collins

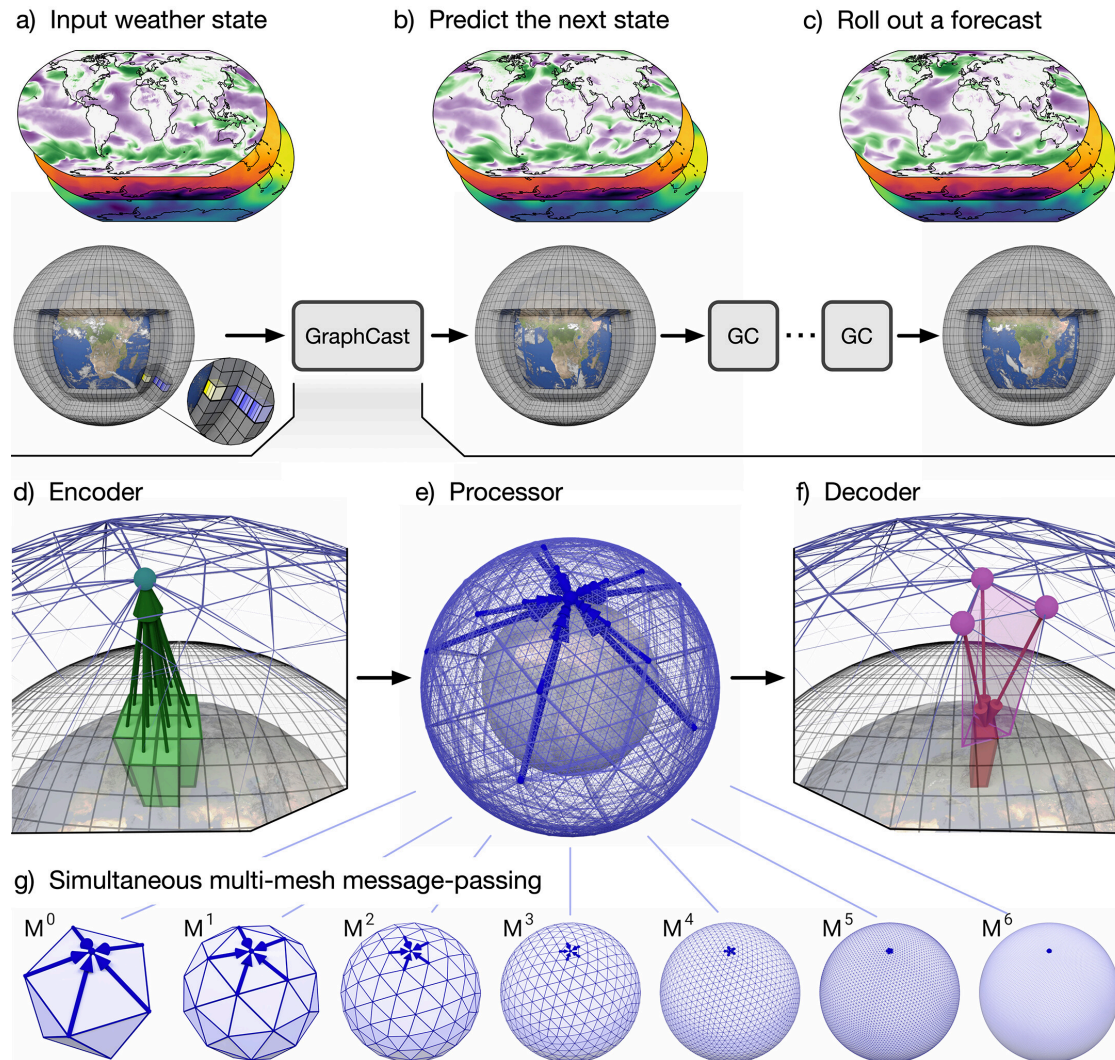
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In Brief

A trained deep neural network predicts antibiotic activity in molecules that are structurally different from known antibiotics, among which Halicin exhibits efficacy against broad-spectrum bacterial infections in mice.

GraphCast: fast and accurate weather forecasts



From neural networks to agents

The breakthrough

Attention Is All You Need

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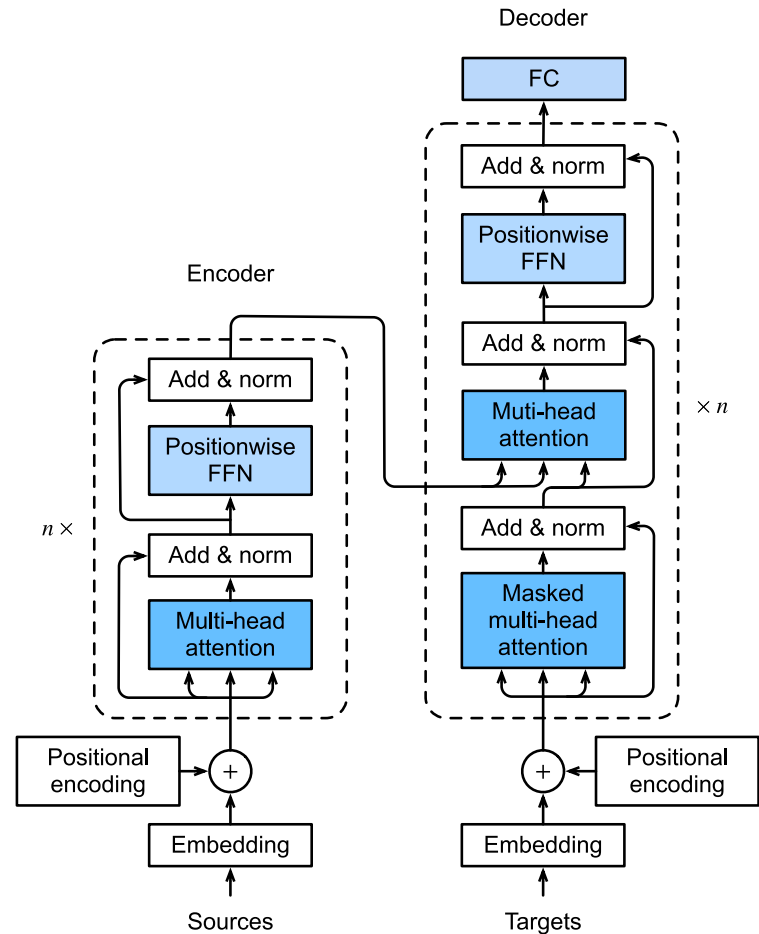
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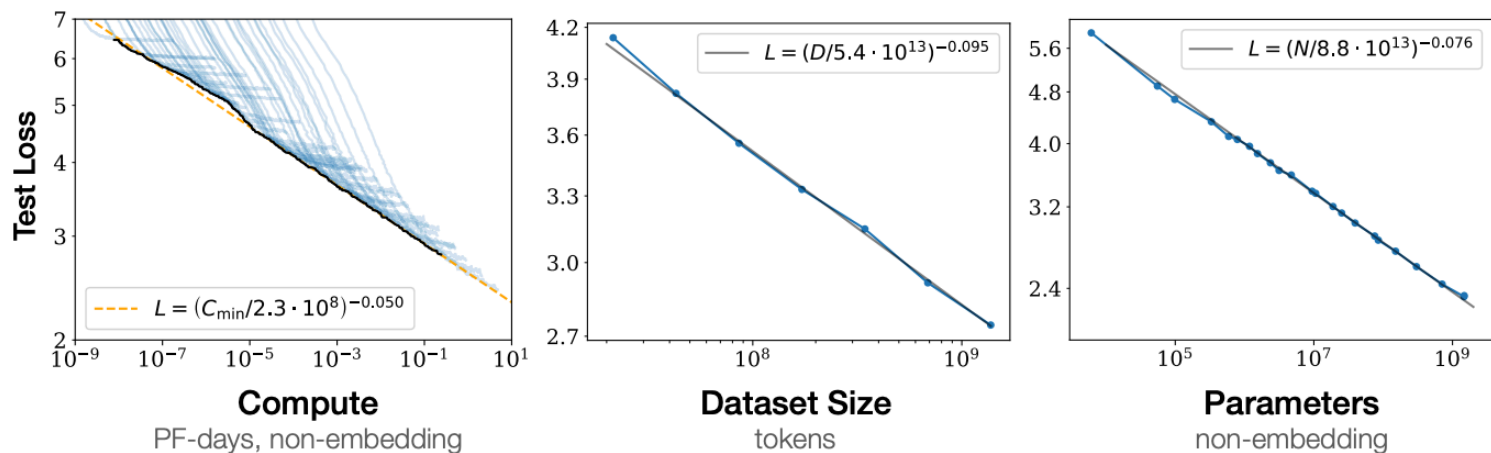
Illia Polosukhin*[‡]
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Vaswani et al., 2017.





A brutal simplicity:

- The more data, the better the model.
- The more parameters, the better the model.
- The more compute, the better the model.

Scaling up further to gigantic models, datasets, and compute resources keeps pushing the boundaries of what is possible, **with no sign of slowing down***.



A role-playing transformer

The same next-word predictor, with a different prompt:

A helpful assistant is having a conversation with a user.

[User]: What should I visit in Liège?

[Assistant]: ____

The model continues the document. Out comes the assistant.



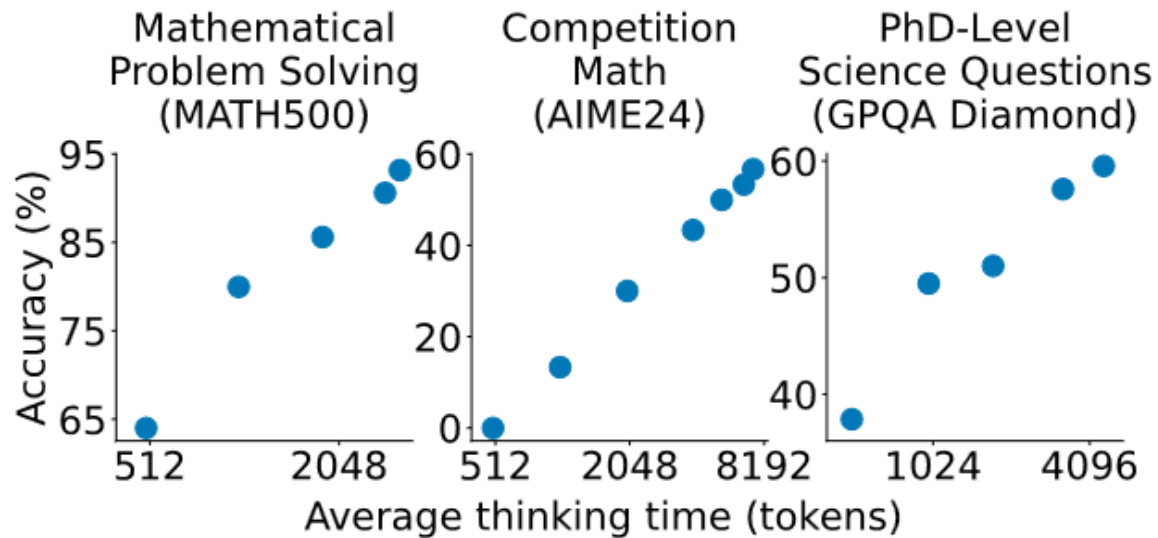
Reasoning by writing

As you would scribble on a whiteboard before answering a hard question, reasoning models **write their thoughts down** before answering.

[User]: How many times does the letter r appear in "strawberry"?

[Assistant, *thinking*]: Let me spell it out: s, t, r, a, w, b, e, r, r, y. The letter r is at positions 3, 8 and 9. [continues for as long as needed]

[Assistant]: Three times.

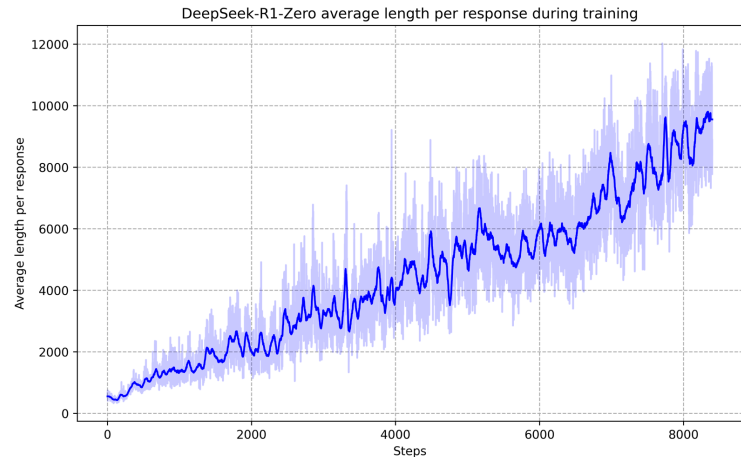


Test-time scaling: letting the model think longer improves its answers.

Learning to reason

Nobody writes the reasoning for the model. It is learned by **reinforcement learning**: the model tries, a program checks the final answer, and successful attempts are reinforced.

Over training, the model learns by itself to think longer, to check its work and to change strategy.



From assistant to agent

If a model can **think** by writing, it can also **act** by writing.

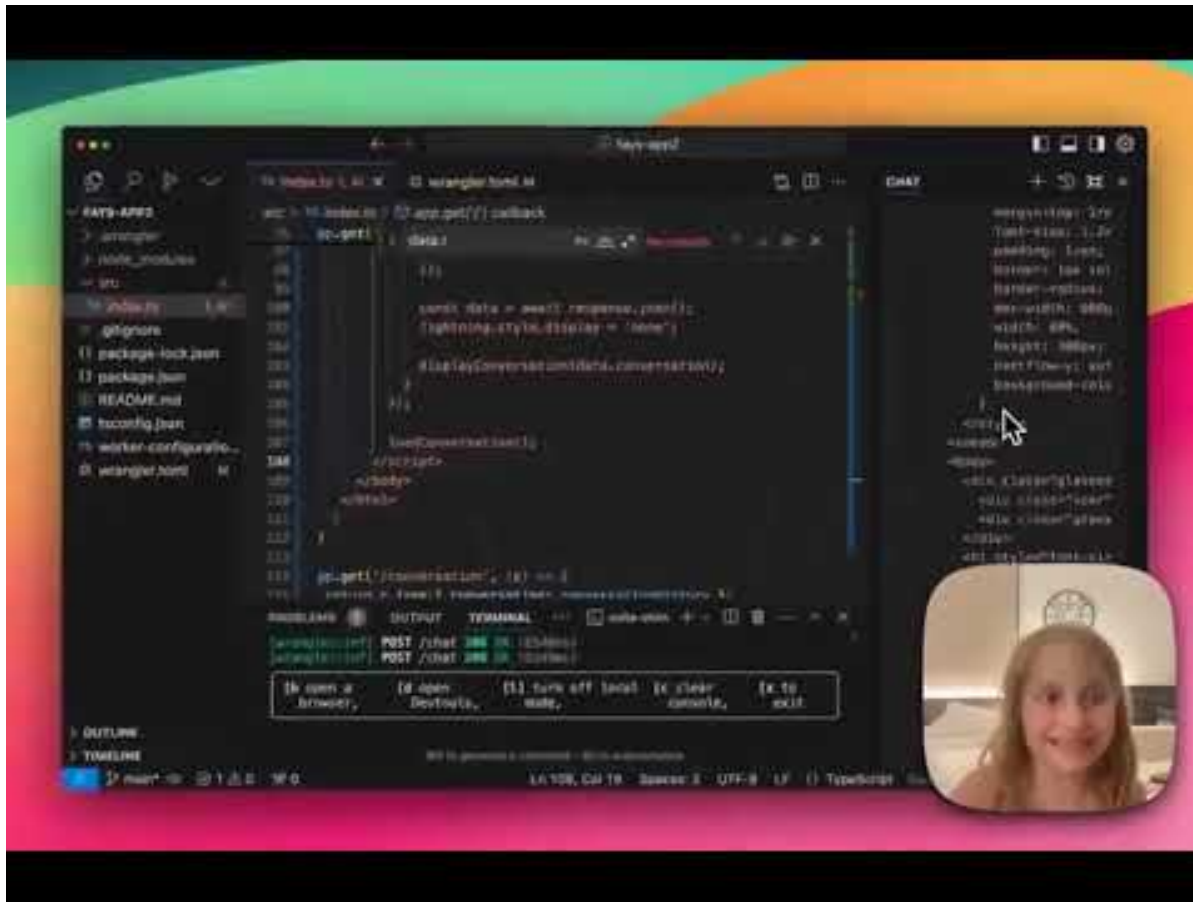
```
[Assistant]: Let me run the tests.
```

```
[Tool call]: bash("pytest tests/")
```

```
[Tool result]: 41 passed, 2 failed:  
test_search.py::test_astar ...
```

```
[Assistant]: The heuristic in astar() is not admissible.  
Let me fix it.
```

A transformer in a loop, whose **tool calls** are intercepted, executed, and fed back into the conversation, is an **agent**. The program that runs this loop is its **harness**.



Code assistants (Cursor; Electroscope, 2024)



Not just text, but also images and sounds (OpenAI, 2024)



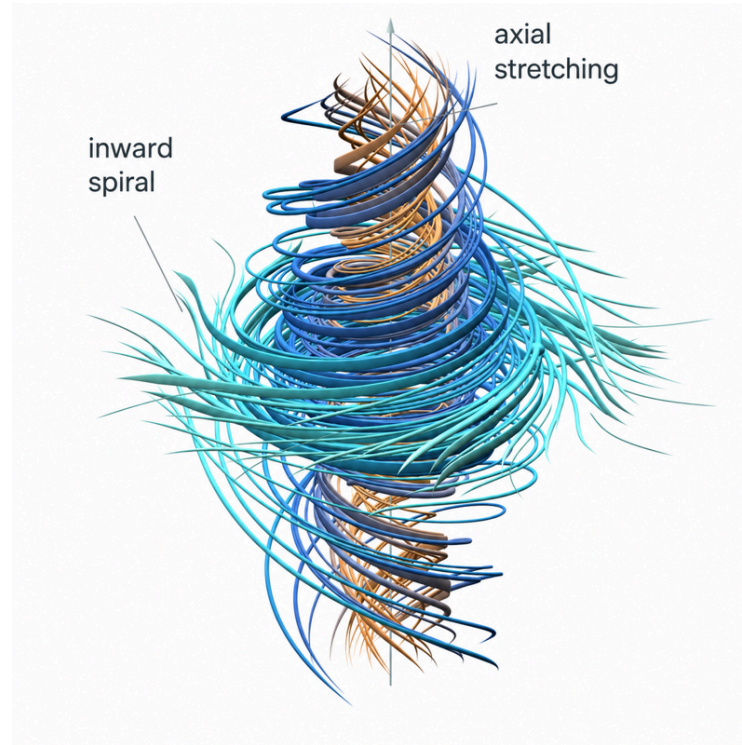
Agents that use computers (OpenAI, GPT-6 Astra, 2026)

Swarms

One agent works in sequence. A **swarm** runs many agents in parallel:

- an orchestrator splits the problem and assigns sub-tasks;
- groups of agents explore different approaches and share intermediate results;
- verifiers check what comes back;
- the best results are merged, and the loop continues.

Computation at inference now scales in two directions: **longer** (thinking) and **wider** (more agents).



The Navier–Stokes problem. Can a fluid that starts smooth reach **infinite speed** in finite time? A Millennium Prize Problem, open for about 90 years.

On September 8, 2026, OpenAI claimed a proof using a swarm of 10,000 agents running in parallel for 88 hours. It produced a formal proof checked by Lean.

Now what?



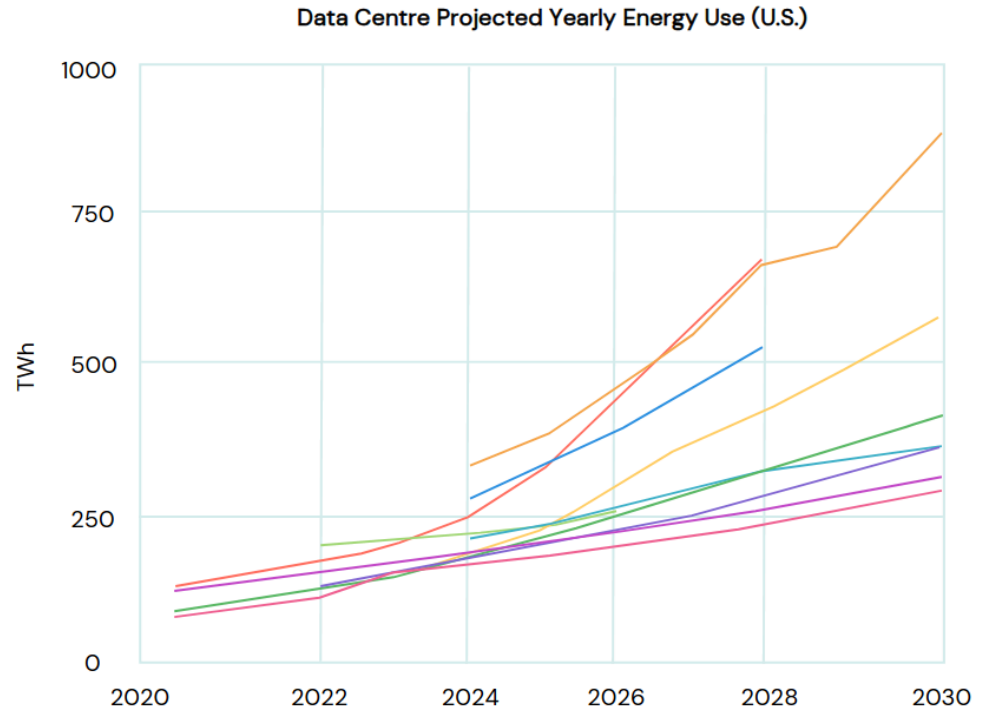
The good

- AI is a tool capable of automating tedious tasks.
- AI can solve complex tasks that no one could solve before.
- The human-machine interface is changing radically, with conversational assistants making digital tools more accessible.
- Agents carry out long, multi-step tasks, from software engineering to mathematics.
- Progress continues at a breakneck pace.



The bad

- Deep learning models are difficult to inspect, debug and explain.
- AI systems can fail in unexpected and catastrophic ways, despite apparent precision.
- Agents act on the world: their mistakes compound over long tasks, and their capabilities (e.g., in cybersecurity) can be misused.
- AI generates a lot of low-quality content ("AI slop") that pollutes information and devalues human creative work.
- Deep learning models have become very large and require significant computational resources, with significant environmental consequences.



"General-purpose AI is a moderate but rapidly growing contributor to global environmental impacts through energy use and greenhouse gas (GHG) emissions. Current estimates indicate that **data centres and data transmission account for an estimated 1% of global energy-related GHG emissions, with AI consuming 10–28% of data centre energy capacity**. AI energy demand is expected to grow substantially [...]"



The ugly

- AI models are biased and can perpetuate discrimination.
- Malicious uses of AI are becoming more frequent (deep fakes, bots, manipulation, etc).
- In all areas of society, a dependence on AI is settling in, with risks of dehumanization, loss of control, and loss of cognitive skills.

Updated: JUN 23, 2025 4:16 PM CET

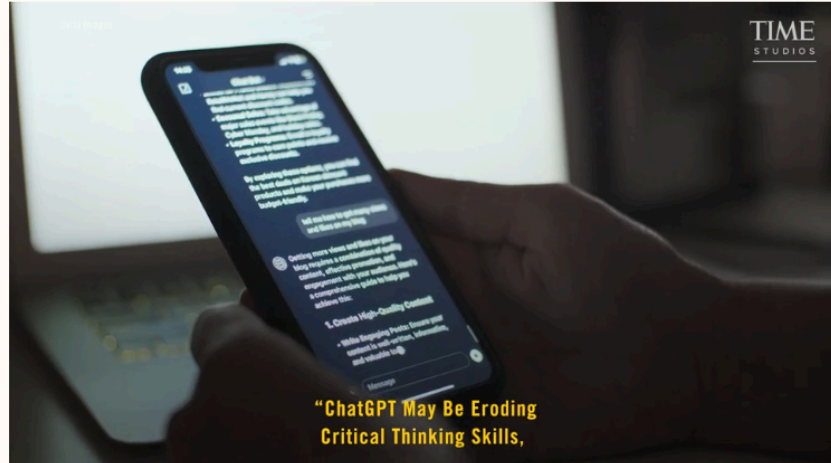
ChatGPT May Be Eroding Critical Thinking Skills, According to a New MIT Study

TECH

ARTIFICIAL INTELLIGENCE



by **Andrew R. Chow**
CORRESPONDENT



TIME
STUDIOS

"Students who used ChatGPT while completing SAT-style essays showed the lowest levels of brain activity. Furthermore, their writing became increasingly formulaic, forgettable, and lacking original thought. Over time, students became increasingly passive and disengaged. Many couldn't recall what they'd written or revise their work without AI support, proof that **they weren't truly learning.**"



That's the thing nobody
warns you about.

Not that it will fail.
That it won't.

Atrophy, a novella

glouppe.github.io/atrophy

INFO8006 Introduction to AI

Why this course?

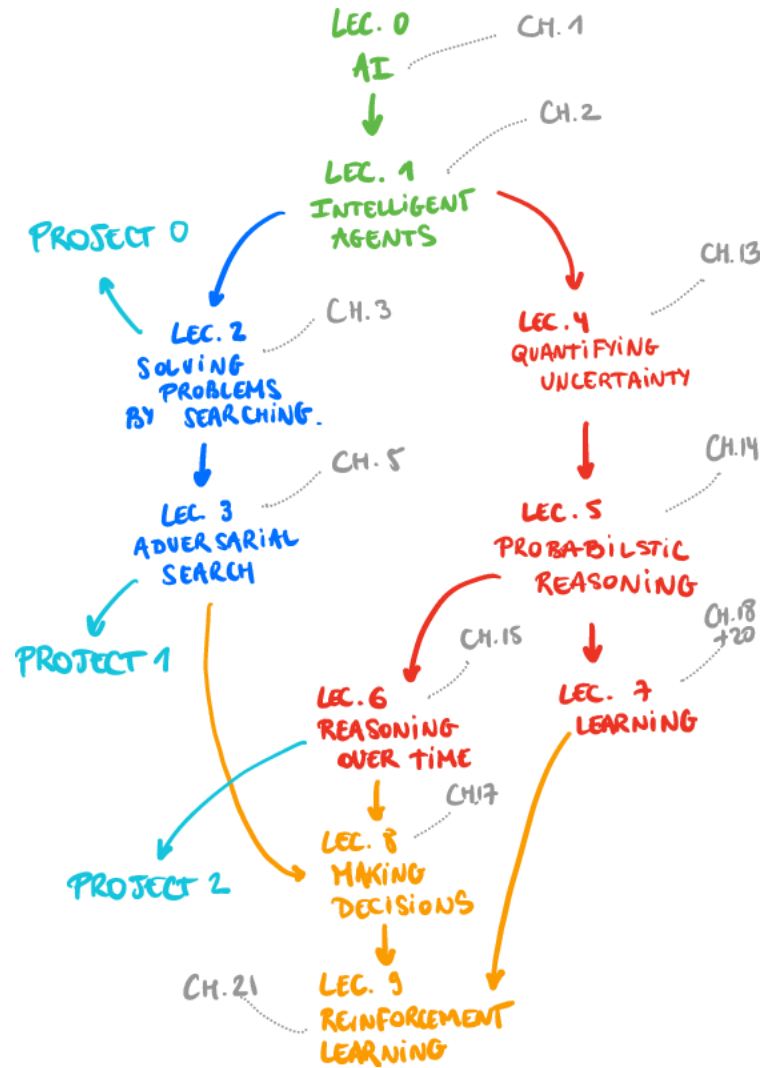
The systems of the previous slides are built on the ideas of this course:

- an agent in a loop with its environment (Lecture 1);
- reasoning as search, and planning against an adversary (Lectures 2 and 3);
- beliefs under uncertainty, and over time (Lectures 4 to 6);
- neural networks and learning from data (Lecture 7);
- decisions and reinforcement learning, which teaches models to reason (Lectures 8 and 9).

We build these ideas from first principles, on problems small enough to understand exactly why an agent behaves as it does.

Course outline

- Lecture 0: Artificial intelligence
- Lecture 1: Intelligent agents
- Lecture 2: Solving problems by searching
- Lecture 3: Games and adversarial search
- Lecture 4: Quantifying uncertainty
- Lecture 5: Probabilistic reasoning
- Lecture 6: Reasoning over time
- Lecture 7: Machine learning and neural networks
- Lecture 8: Making decisions
- Lecture 9: Reinforcement learning



My mission

By the end of this course, you will have built autonomous agents that efficiently make decisions in fully informed, partially observable and adversarial settings. Your agents will draw inferences in uncertain and unknown environments and optimize actions for arbitrary reward structures.

The models and algorithms you will learn in this course apply to a wide variety of artificial intelligence problems and will serve as the foundation for further study in any application area (from engineering and science, to business and medicine) you choose to pursue.

Going further

This course is designed as an introduction to the many other courses available at ULiège and (broadly) related to AI, including:

- INFO8006: Introduction to Artificial Intelligence ← you are there
- DATS0001: Foundations of Data Science
- ELEN0062: Introduction to Machine Learning
- INFO8010: Deep Learning
- INFO8004: Advanced Machine Learning
- INFO9023: Machine Learning Systems Design
- INFO8003: Reinforcement learning
- INFO0948: Introduction to Intelligent Robotics
- INFO9014: Knowledge representation and reasoning
- ELEN0016: Computer vision

The end.